

NexusERP Operations Portal

Technical Documentation & Architecture Specification

1. Executive Summary

NexusERP Operations Portal is a full-stack Mini ERP & CRM system engineered for wholesale and distribution companies. It streamlines customer relationship management (CRM), stock inventory management with real-time alerts and audit logs, and multi-item sales challan processing with atomic stock reduction and PDF invoice generation.

2. Live System Links

- GitHub Repository: <https://github.com/nithishdabbara/NexusERP-Operations-Portal>
- Live Frontend Web App: <https://nexus-erp-operations-portal.vercel.app>
- Live Backend REST API: <https://nexuserp-backend-ja9d.onrender.com/api>
- Live Cloud Database: Supabase PostgreSQL (ijnnazbvnufevfyappts)
- Docker Hub Images: [nithish1875/nexuserp-backend:latest](#) & [nithish1875/nexuserp-frontend:latest](#)

3. Test Login Credentials (Password: Password123!)

- Admin Role: admin@company.com — Full access across all ERP & CRM modules
- Sales Role: sales@company.com — Customer CRM, Follow-ups, Create/Confirm Sales Challans
- Warehouse Role: warehouse@company.com — Manage Products, Stock Level Adjustments (IN/OUT)
- Accounts Role: accounts@company.com — View Challans, Customer Lists, Financial Dashboard

4. System Architecture

- Frontend Layer: React 18, TypeScript, Vite, responsive dark glassmorphic UI with 1-Click Role Switcher.
- Backend Layer: Node.js, Express.js, TypeScript REST APIs with Zod schema validation & error handling.
- Database Layer: PostgreSQL hosted on Supabase, managed via Prisma ORM for schema type-safety.
- Stock Transaction Engine: Executes atomic stock checks & reductions via Prisma \$transaction.
- Streaming Invoice Generator: Dynamic PDF invoice rendering using pdfkit.

5. Core REST API Endpoints

- POST /api/auth/login — Authenticate user and issue JWT token
- GET /api/customers — List customers with search, type, and status filtering
- POST /api/customers/:id/followups — Add CRM follow-up note
- GET /api/products — Product catalog and minimum stock alert list
- POST /api/products/:id/adjust-stock — Manual stock adjustment (IN/OUT)
- GET /api/products/stock-logs — Global stock movement audit history
- POST /api/challans — Create sales challan (Draft or Confirmed with stock reduction)
- GET /api/challans/:id/pdf — Stream & download printable PDF Invoice

6. Known Limitations & System Scope

- Render Cold Start: Free tier backend service sleeps after inactivity; first request takes ~30s to spin up.
- Single Company Context: Built specifically for single-tenant internal distribution operations.